REINVENT COMMERCE / TRACK 03

Rhythm

A personal purchase clock for customer retention

Notice when a regular shopper’s routine changes. Give a merchant evidence for a thoughtful next step.

Prototype by KathanSena

Repeat buying has a personal rhythm

THE MERCHANT

A small retailer has order history but limited time to inspect every repeat shopper.

THE GAP

A fixed “inactive for 30 days” rule treats weekly buyers and seasonal buyers alike.

Rhythm asks a narrower question: has this shopper gone beyond their own observed repeat pattern?

From transaction history to a reviewed action

1 Find a changed cadence

Analyze a sales CSV in the browser. Rank repeat items by overdue shopper count.

2 Inspect the evidence

Review the shopper's last order, usual gap, time elapsed, and most repeated item. IDs stay masked in the interface.

3 Draft with a human

Optionally ask Gemma for a low pressure reorder reminder. A person reviews it. Rhythm never sends messages.

The model writes wording. A deterministic rule creates the review signal.

A transparent personal cadence rule

ELIGIBILITY

At least four distinct purchase days

THRESHOLD

max(14 days, 1.6 × median gap)

REVIEW SIGNAL

The interval changed. Merchant context helps explain why.

Flag for human review only when days since the last order exceed that threshold.

A real dataset, with clear limits

UCI ONLINE RETAIL

A reproducible extract of UK transactions from 2010–2011

The prototype loads research data, builds shopper and item queues, shows purchase evidence, and accepts merchant CSVs.

WHAT IT SHOWS

Personal cadence flags, item level counts, and evidence for merchant review.

EVIDENCE LIMITS

Historical transactions; live merchant results and sales impact remain unmeasured.

Source: UCI Machine Learning Repository · Online Retail · Chen (2015) · DOI 10.24432/C5BW33 · CC BY 4.0

Item watchlist for merchant review

ITEM SIGNAL

Rank repeat items with overdue shoppers.

Show the typical repeat interval and median lateness. Open matching shoppers to inspect their purchase evidence.

CURRENT STATUS

A practical shortlist from historical purchase patterns. Merchants decide which shoppers merit follow-up.

TECHNOLOGY

Next.js · TypeScript · browser-side CSV analysis · Gemma through LM Studio